

IF

What does it do?

It checks whether a condition is true or not and on the basis of that returns a value

Syntax

=if(Logical_Test, [value_if_true], [value_if_false])

Logical_Test

This will be the condition which you want to check, so greater than, less than or equal to signs can be used for numbers. If you want to evaluate a condition based on **text value**, then use double quotes (") with equal to signs (Row 27 for example)

Value_if_True

If the logical test holds, then excel will return this value

Value_if_False

If the logical test does not hold, then excel will return this value

Example

Using IF formula to identify if a student has passed an exam or not

Student ID	Marks	Pass or Fail	Formula
S001	90	Pass	=IF(B22>=40,"Pass","Fail")
S002	70	Pass	=IF(B23>=40,"Pass","Fail")
S003	20	Fail	=IF(B24>=40,"Pass","Fail")
S004	41	Pass	=IF(B25>=40,"Pass","Fail")

Note

If instead of marks, we had Grades A, B, C i.e. **text values** then the above formulas would have changed by IF(B21= 'A',"Pass","Fail")

Exercise

Write a formula using IF. Salary for an AVP will be \$2,000 and for an analyst will be \$1,000

Emp ID	Designation	Salary
E001	AVP	\$2,000
E002	Analyst	\$1,000
E003	Analyst	\$1,000
E004	Analyst	\$1,000



umbers. When you want to

AND

What does it do?

Checks if all arguments are true and returns TRUE value else returns FALSE value. You need to have minimum 1 argument. This function is used with IF condition a lot

Syntax

=AND(Logical 1, Logical 2, Logical 3, and so on..)

Logical 1

First condition which needs to be checked

Logical 2

Second condition which needs to be checked

Example 1

Testing condition 1 and condition 2 to check the numbers are greater than 0 using AND

Condition 1	Condition 2	Result	Formula
1	2	TRUE	=AND(A19>0,B19>0)
1	-2	FALSE	=AND(A20>0,B20>0)

Example 2

Use IF with AND function to check if student has passed or not. If the student has failed in 1 of the 2 subjects, the student is considered as Failed

Student ID	Subject 1	Subject 2	Overall Result	Formula
S001	PASS	PASS	PASS	=IF(AND(B26="PASS",C26="PASS"),'PASS','FAIL')
S002	PASS	FAIL	FAIL	=IF(AND(B27="PASS",C27="PASS"),'PASS','FAIL')
S003	FAIL	PASS	FAIL	=IF(AND(B28="PASS",C28="PASS"),'PASS','FAIL')
S004	PASS	PASS	PASS	=IF(AND(B29="PASS",C29="PASS"),'PASS','FAIL')

Exercise

Write a formula using AND with IF to find if an employee should get a "GOOD" increment or not. Sales must be more than 50 and he should have adhered to all HR policies

Emp ID	Designation	Sales	Adhered to HR Policies	GOOD Increment (Yes or NO)
E001	AVP	51	YES	Yes
E002	Analyst	100	NO	No
E003	Analyst	49	YES	No
E004	Analyst	75	YES	Yes



um 2 conditions to check. AND

ts, then the student will be

"PASS", "FAIL")
"PASS", "FAIL")
"PASS", "FAIL")
"PASS", "FAIL")

le by the employee should be

Nested IF's

What does it do?

Nested IF conditions help in checking multiple conditions together by using IF conditions within IF condition.

Syntax

=if(Logical_Test, [value_if_true], if(Logical_Test, [value_if_true], [value_if_false]))

Logical_Test

This will be the condition which you want to check, so greater than, less than or equal to signs can be used for condition based on **text value**, then use double quotes (") with equal to signs

Value_if_True

If the logical test holds, then excel will return this value

Value_if_False

If the logical test does not hold, then excel will return this value

Nested IF

From Excel 2007 onwards, we can use 64 IF conditions in one formula whereas Excel 2003 allows for only 7 IF

Example

Use Nested IF to put the grade a student gets in an exam in Table B. Score Ranges for different grades are given in Table A.

TABLE A	
Score	Grade
> 70	A
41-70	B
< 40	C

TABLE B			
Student ID	Marks	Grade	Formula
S001	90	A	=IF(E26< 41,"C",IF(E26
S002	70	B	=IF(E27< 41,"C",IF(E27
S003	20	C	=IF(E28< 41,"C",IF(E28
S004	41	B	=IF(E29< 41,"C",IF(E29

Exercise

Write a formula using Nested IF conditions to check what level increment an employee should get and then display the result in dollars using Format

TABLE A	
Performance	Increment
A	\$1,000
B	\$500
C	\$250

TABLE B			
Emp ID	Designation	Performance	Result
E001	AVP	A	\$2,000
E002	Analyst	A	\$250
E003	Analyst	B	\$250
E004	Analyst	C	\$250



or numbers. When you want to evaluate a

= conditions

ven in Table A.

< 71,"B","A"))
< 71,"B","A"))
< 71,"B","A"))
< 71,"B","A"))

convert the amount into

OR

What does it do?

Checks if any of the conditions are true or not. It returns TRUE value even if one of the conditions holds else a FALSE to appear, no conditions should hold. You need to have minimum 2 conditions to check. AND function

Syntax

=OR(Logical 1, Logical 2, Logical 3, and so on..)

Logical 1

First condition which needs to be checked

Logical 2

Second condition which needs to be checked

Example 1

Testing condition 1 and condition 2 to check the numbers are greater than 0 using OR

Condition 1	Condition 2	Result	Formula
1	2	TRUE	=OR(A20>0,B20>0)
1	-2	TRUE	=OR(A21>0,B21>0)
-1	-2	FALSE	=OR(A22>0,B22>0)

Example 2

Use IF with OR function to check if student has passed or not. If the student has failed in 1 of the 2 subjects considered as Failed

Student ID	Subject 1	Subject 2	Overall Result	Formula
S001	PASS	PASS	PASS	=IF(OR(B28="FAIL",C28="FAIL"),"F",P)
S002	PASS	FAIL	FAIL	=IF(OR(B29="FAIL",C29="FAIL"),"F",P)
S003	FAIL	PASS	FAIL	=IF(OR(B30="FAIL",C30="FAIL"),"F",P)
S004	PASS	PASS	PASS	=IF(OR(B31="FAIL",C31="FAIL"),"F",P)

Exercise

Write a formula using OR with IF to find if an employee should get a "GOOD" increment or not. Either Sales more than 50 or he should have adhered to all HR policies

Emp ID	Designation	Sales	Adhered to HR Policies	GOOD Increment (Yes or NO)
E001	AVP	51	YES	Yes
E002	Analyst	49	NO	NO
E003	Analyst	49	YES	Yes
E004	Analyst	75	YES	Yes



e returns FALSE value. Basically, for
tion is used with IF condition a lot

;, then the student will be

:AIL", "PASS")
:AIL", "PASS")
:AIL", "PASS")
:AIL", "PASS")

; made by the employee should be

IFERROR

What does it do?

Sometimes when we use formula in cells, we might get an error because of the cell value, IFERROR allows us to handle the error and return the value we want

Syntax

=IFERROR(Value,[Value_if_error])

Logical 1

First cell value which is checked if there is an error

Value_if_error

If in the cell there is an error, that error value will be replaced by this value

Example 1

Divide Col A by Col B and show the results without using IFERROR and with using IFERROR

Col A	Col B	Without IFERROR &Formula		With IFERROR &Formula	
1	2	0.5	=A19/B19	0.5	=IFERROR(A19/B19,999)
1	0	#DIV/0!	=A20/B20	999	=IFERROR(A20/B20,999)

Note

In the above example, on dividing 1 by 0, we get an error, without using IFERROR we get #DIV/0! i.e. an error value which is not the value we want. You can also enter text which can let the user know that there is some problem with the data.

Example 2

Divide Col A by Col B and show the results without using IFERROR and with using IFERROR

Col A	Col B	Without IFERROR &Formula		With IFERROR &Formula	
1	2	2	=A29*B29	2	=IFERROR(A29*B29,999)
1	1	1	=A30*B30	1	=IFERROR(A30*B30,999)
1	a	#VALUE!	=A31*B31	DATA IS WRONG	=IFERROR(A31*B31,999)
5	a	#VALUE!	=A32*B32	DATA IS WRONG	=IFERROR(A32*B32,999)

Exercise

Write a formula using IFERROR. Find the monthly salary of each employee using data in Column C and if an error occurs, return "ERROR IN SALARY". Change the format to £ (Pounds) and no decimal places

Emp ID	Designation	Annual Salary	Monthly Salary
E001	AVP	£80,000	£ 6,667
E002	Analyst	£50,000	£ 4,167
E003	Analyst	abcd	ERROR IN SALARY
E004	Analyst	£40,000	£ 3,333



is to remove such errors with values

)
)

or. So using IFERROR, we can give
ata.

ila
"DATA IS WRONG")
"DATA IS WRONG")
"DATA IS WRONG")
"DATA IS WRONG")

The red color highlight is for
representation here - color cannot
be changed using IFERROR

error appears, then change it to

IFNA

What does it do?

Sometimes when we use formula in cells, we get values errors of type #N/A. We can use IFNA to remove the error value. This function is from Excel 2013 onwards, so is not available in earlier versions of Excel.

Syntax

=IFNA(Value,[Value_if_error])

Logical 1

First cell value which is checked if there is an error

Value_if_error

If in the cell there is a value of #N/A, the error value will be replaced by this value

Example 1

Divide Col A by Col B and show the results without using IFERROR and with using IFERROR

Col A	With IFERROR & Formula	
1	1	=IFNA(A19,9999)
#N/A	9999	=IFNA(A20,9999)

The red color highlighting is used to represent an error value. This can be changed using the IFNA function.

Exercise

Write a formula using IFNA. Find the monthly salary of each employee using data in Column C and if #N/A is returned, display "ERROR IN SALARY". Change the format to £ (Pounds) and no decimal places

Emp ID	Designation	Annual Salary	Monthly Salary
E001	AVP	£80,000	£ 6,667
E002	Analyst	£50,000	£ 4,167
E003	Analyst	#N/A	ERROR IN SALARY
E004	Analyst	£40,000	£ 3,333



is with our choice of value. IFNA

ght is for
olor cannot
IFNA

appears, then change it to ERROR IN

EXACT

What does it do?

It checks whether 2 values (texts, numbers) are exactly same or not. If they are same, then it will return TRUE. If not, it will return FALSE. This function is case-sensitive.

Syntax

=EXACT(Text1, Text2)

Text1

First string which you want to check

Text2

Second string which you want to test against

Example 1

Testing condition 1 and condition 2 to check the numbers are greater than 0 using AND

Col A	Col B	Result	Formula
EXCEL	EXCEL	TRUE	=EXACT(A19,B19)
EXCEL	Excel	FALSE	=EXACT(A20,B20)

Note

In the above example, Row 20 result is FALSE because the EXACT function is case-sensitive. The first value is EXCEL and the second one (Excel) is a combination of upper and lower capse.

Exercise

Write a formula using EXACT to check if the values in Col A and Col B are same or not

Col A	Col B	Results
TEST	Test	TRUE
TEST	TEST	FALSE
1234	1234	FALSE
1.1	1.101	TRUE



!UE else it will return FALSE. This

s in upper capse (EXCEL) whereas

TEXT

What does it do?

It convert a numeric value into text based on the format given by the user

Syntax

=TEXT(Value, Format_text)

Value

Value for which we need to change the format

Format_text

Format specified by the user

Example 1

Using different text formats

Col A	Result	Formula
27/03/2017	27/03/2017	=TEXT(A18, "dd/mm/yyyy")
42821	27/3/2017	=TEXT(A19, "dd/m/yyyy")
42821	Mar 27 2017	=TEXT(A20, "mmm dd yyyy")
27/03/2017	Mar-17	=TEXT(A21, "mmm-yy")
14.141414	14.14	=TEXT(A22, "0.00")
14598.12	\$14,598.12	=TEXT(A23, "\$#,##0.00")
14598.12	£14,598.12	=TEXT(A24, "£#,##0.00")
14:00	14:00	=TEXT(A25, "hh:mm")

Using Text function the
appear in the same way
changed

Note:

Col A Row 19 & 20 - Excel stores dates in numbers, so that is why when we use Text with Date formats, Excel cor

Exercise

Write a formula using EXACT to check if the values in Col A and Col B are same or not

Value	Convert To	Results
21/05/2017	21/05/2017	TRUE
42876	21/05/2017	TRUE
12:00	12:00	TRUE
12345.78	£12,345.78	TRUE
12345.78	£12,346	FALSE



ough the results may
; but there format has
to text.

verts those numbers dates

CONCATENATE

What does it do?

This function joins several (more than 1) text string into 1 string

Syntax

=CONCATENATE(String1, String2, and so on..)

OR

=String1&String2& so on..

String 1

First string which you want to concatenate

String 2

Second string which you want to join to the first string

Example

Join the 3 strings together using CONCATENATE and & (ampersand) operator

String1	String2	String3	Result	Formula
S001	90	Pass	S00190Pass	=CONCATENATE(A20,B20,C20)
S002	70	Pass	S00270Pass	=CONCATENATE(A21,B21,C21)
S003	20	Fail	S00320Fail	=A22&B22&C22
S004	41	Pass	S00441Pass	=A23&B23&C23
S005	39	Fail	S005 39 Fail	=CONCATENATE(A24," ",B24," ",C24)
S006	99	Pass	S006 99 Pass	=A25&" "&B25&" "&C25
S007	11	Fail	S007;11;Fail	=CONCATENATE(A26,";",B26,";",C26)

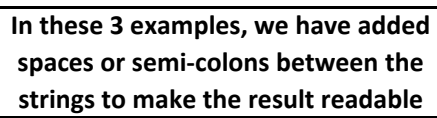
Note

In order to add spaces, other special characters between strings, you would need to use double quotes "" as done in the examples above

Exercise

Join the 3 strings together using CONCATENATE and & (ampersand) operator to get the results as mentioned

String1	String2	String3	Result	USE	Formula
E001	AVP	800	E001-AVP-800	CONCATENATE	E001-AVP-800
E002	Analyst	500	E002Analyst500	CONCATENATE	E002Analyst500
E003	Analyst	100	E003/Analyst-100	CONCATENATE	E003Analyst100
E004	Analyst	400	E004:Analyst:400	&	E004 Analyst 400
E005	AVP	1000	E005 AVP 1000	&	E005 AVP 1000
E006	Analyst	250	E006 Analyst 250	&	E006 Analyst 250
E007	Analyst	500	E007;Analyst*500	&	E007 Analyst 500



TRIM

What does it do?

The function removes spaces all before and following the words i.e. leading and trailing spaces - and also removes multiple spaces between words. TRIM function is used to clean data.

Syntax

=TRIM(Text)

Text

First string which you want to concatenate

Example

Use TRIM to remove spaces

String1	Result	Formula
S001 A	S001 A	=TRIM(A16)
S001 A	S001 A	=TRIM(A17)
S002	S002	=TRIM(A18)
S003	S003	=TRIM(A19)

Exercise

Remove the extra spaces in the strings and then join the 3 strings together using CONCATENATE and & (as mentioned in the result column). Do so in one formula itself

String1	String2	String3	Result	USE	Formula
E001	AVP	800	E001-AVP-800	CONCATENATE	E001-AVP-800
E002	Analyst	500	E002Analyst500	CONCATENATE	E002Analyst500
E003	Ana lyst	100	E003/Analyst-100	&	E003/Ana lyst-100
E004	Analyst	400	E004:Analyst:400	&	E004:Analyst:400



removes extra spaces between words

persand) operator to get the results as

)
)

SUBSTITUTE

What does it do?

This function is used to replace a part of text in a string with something else. SUBSTITUTE function is used to replace text in a string. It is case sensitive.

Syntax

=SUBSTITUTE(Text, Old_Text, New_Text, [instance_number])

Text

String in which you want to modify text

Old_Text

The old text which you want to replace

New Text

The new text with which you want to modify your original text with

Instance_Number

The instance of old text which you want to replace with new text. It is optional but if not provided it will replace all the old text with new text. Look at the examples below for better understanding of this

Example

Join the 3 strings together using CONCATENATE and & (ampersand) operator

String	Instances	Result	Formula
Hut		Hub	=SUBSTITUTE(Hut, "t", "b")
bumble	1	Tumble	=SUBSTITUTE(bumble, "b", "T")
the car is in the driveway		car is in driveway	=SUBSTITUTE(the car is in the driveway, "the ", "")
the car is in the driveway	2	the car is in a driveway	=SUBSTITUTE(the car is in the driveway, "the ", "a ")
The car is in the driveway		car is in the driveway	=SUBSTITUTE(The car is in the driveway, "The ", "")

Note

If instances are not mentioned, then excel will replace all the occurrences as can be seen on Row 28. And the same can be seen on Row 30

Exercise

Use SUBSTITUTE function to make changes to some of the strings below to get the results mentioned in the table

String	Result	Formula
The dog is in the lawn	dog is in the lawn	dog is in the lawn
the dog is in the lawn	Dog is in lawn	the dog is in lawn
Dog	Fog	Fog
Curious Cat	Furious Cat	Curious Cat



to clean data. This function is case-

ake changes to all the occurrences of

Formula
<code>=JTE(A26,"t","b")</code>
<code>=JTE(A27,"b","T",1)</code>
<code>=JTE(A28,"the","")</code>
<code>=JTE(A29,"the","a",2)</code>
<code>=JTE(A30,"The","")</code>

he function is case-sensitive which can

e 'Result' column

VALUE

What does it do?

The function is used to convert text value into numeric value

Syntax

=VALUE(Text)

Text

String which you want to convert to number

Example

Convert the values into numbers

Text	Result	Formula
\$12,345	12345	=VALUE(A15)
27/03/2017	42821	=VALUE(A16)
09/12/2014	41982	=VALUE(A17)

Excel stores dates as numbers, hence it changes the dates to numbers

Exercise

Convert the values in Text column into numbers

Text	Result	Formula
\$123.17	123	123.17
01/12/2017	43070	43070
12/01/2017	42747	42747



he function
rs

UPPER

What does it do?

This function converts the entire text string into uppercase letters

Syntax

=UPPER(Text)

Text

String for which you want to convert all letters into UPGASE

Example

Convert the values in Text column into UPPERCASE

Text	Result	Formula
Laptop	LAPTOP	=UPPER(A15)
excel	EXCEL	=UPPER(A16)
superstore is closed now	SUPERSTORE IS CLOSED NOW	=UPPER(A17)

Exercise 1

Convert the values in Text column into UPPERCASE

Text	Formula
good	GOOD
how are you?	HOW ARE YOU?

Exercise 2

Use UPPER to change Col A and then use EXACT function to match the 2 strings to see if they are same or not

Col A	Col B	Result
MATCH	MATCH	TRUE
MATCH	Match	FALSE
CORECT	CORECT	TRUE



LOWER

What does it do?

This function converts the entire text string into lowercase letters

Syntax

=LOWER(Text)

Text

String for which you want to convert all letters into lowercase

Example

Convert the values in Text column into lowercase

Text	Result	Formula
Laptop	laptop	=LOWER(A15)
EXCEL	excel	=LOWER(A16)
SUPERSTORE IS CLOSED NOW	superstore is closed now	=LOWER(A17)

Exercise 1

Convert the values in Text column into lowercase

Text	Formula
GOOD	good
HOW are you?	how are you?

Exercise 2

Use LOWER to change Col A and then use EXACT function to match the 2 strings to see if they are same or

Col A	Col B	Result
match	match	TRUE
match	mATCH	FALSE
correct	correct	TRUE



not

LEN

What does it do?

This function returns the number of characters in a string (text or numeric value). It also counts the space is used for data manipulation along with LEFT, RIGHT functions

Syntax

=LEN(Text)

Text

String for which you want to find the length

Example

Convert the values in Text column into lowercase

Text	Result	Formula
Laptop	6	=LEN(A16)
EXCEL	6	=LEN(A17)
SUPERSTORE IS CLOSED ;	22	=LEN(A18)
90	2	=LEN(A19)

Exercise 1

Find the length of the followig strings

Text	Formula
Phone Number is +(0)1234	24
This is used for data cleaning	30



s and special characters. This function

LEFT

What does it do?

This function returns the number of specified characters from the start of the string. This function is used for remembering that you read from left to right - so that is why from the start of the string

Syntax

=LEFT(Text, [num_chars])

Text

String from which you want to find letters

num_chars

Number of characters you want to read from the left

Example

Use LEFT function on the following strings

Text	Result	Formula
Laptop	Lapto	=LEFT(A19,5)
EXCEL IS WOW	EXCEL IS	=LEFT(A20,8)
SUPERSTORE IS CLOSED ;	SUPERSTORE IS C	=LEFT(A21,15)
90	9	=LEFT(A22,1)
SUPERSTORE IS CLOSED ;	SUPERSTORE IS C	=LEFT(A23,LEN(A23)-7)

Exercise 1

Use LEFT function to get the results in the 2nd column

Text	Result	Formula
Phone Number is +(0)1234	Phone Number is +(0	Phone Number is +(0
This is used for data cleaning	This is used for data cle	This is used for data cle

Exercise 2

Use LEFT and LEN function together to only remove last 5 characters of the string

Text	Result	Formula
Phone Number is +(0)1234	Phone Number is +(0	Phone Number is +(0
This is used for data cleaning	This is used for data cle	This is used for data cle



or data manipulation. A way to

RIGHT

What does it do?

This function returns the number of specified characters from the end of the string. This function is used for data that you read from left to right - so that is why from the end of the string

Syntax

=RIGHT(Text, [num_chars])

Text

String from which you want to find letters

num_chars

Number of characters you want to read from the left

Example

Use RIGHT function on the following strings

Text	Result	Formula
Laptop	top	=RIGHT(A19,3)
EXCEL IS WOW	WOW	=RIGHT(A20,4)
SUPERSTORE IS CLOSED ;	OSD ;	=RIGHT(A21,6)
90	0	=RIGHT(A22,1)
SUPERSTORE IS CLOSED ;	OSD ;	=RIGHT(A23,LEN(A23)-16)

Exercise 1

Use RIGHT function to get the results in the 2nd column

Text	Result	Formula
Phone Number is +(0)1234	is +(0)1234	is +(0)1234
This is used for data cleaning	data cleaning	data cleaning

Exercise 2

Use RIGHT and LEN function together to only remove first 5 characters of the string

Text	Result	Formula
Phone Number is +(0)1234	Number is +(0)1234	Number is +(0)1234
This is used for data cleaning	is used for data cleaning	is used for data cleaning



a manipulation. A way to remember is

MID

What does it do?

This function returns the number of specified characters from the left to right but, you can change the starting position different from LEFT function.

Syntax

=MID(Text, [starting_position], [num_chars])

Text

String from which you want to find letters

starting_position

Starting position of the string

num_chars

Number of characters you want to read from the starting of the string

Example

Use RIGHT function on the following strings

Text	Result	Formula
Laptop restarted	top re	=MID(A22,4,6)
EXCEL IS WOW	XCEL IS W	=MID(A23,2,9)
SUPERSTORE IS CLOSED ;	E IS CLOS	=MID(A24,10,9)
90 EMPLOYEES	0 EMPL	=MID(A25,2,6)

Exercise

Use MID function to get the results in the 2nd column

Text	Result	Formula
Phone Number is +(0)1234	is +(0)1	is +(0)1
This is used for data cleaning	for data clean	for data clean
How have you been?	you	you
Any questions?	quest	quest



art of the string thus making MID

DATE

What does it do?

This function returns the date when days, month, year are stored in different cells. If the cell in which the formula is entered then the function will return a number - excel stores dates as numbers

Syntax

=DATE(Year,Month,Day)

Year

Year of the date. Try to supply the entire 4 digits

Month

Month of the date in numeric format like for January it would be 1, for February it would be 2

Day

Day of the date - it needs to be in numeric format like 1, 2, 3 ...and till 31

Example

Use DATE function to calculate dates

Year	Month	Day	Result	Formula
2017	12	31	31/12/2017	=DATE(A22,B22,C22)
2016	11	30	30/11/2016	=DATE(A23,B23,C23)
2015	3	1	01/03/2015	=DATE(A24,B24,C24)
2014	8	1	01/08/2014	=DATE(A25,B25,C25)

Exercise 1

Use DATE function to calculate dates and then format the result into Date format

Year	Month	Day	Formula
2012	7	31	31/07/2012
2011	1	30	30/01/2011
2010	2	1	01/02/2010
2000	5	25	25/05/2000

Exercise 2

Use LEFT, MID and RIGHT functions to extract day, month and year from Col A and then use DATE function to

Col A	Year	Month	Day	Date
21122016	2016	12	15	15/12/2016
13122000	2000	12	10	10/12/2000
15011999	1999	01	15	15/01/1999
10102010	2010	10	10	10/10/2010



action is being called is of Number format,

› calculate dates

DATEVALUE

What does it do?

This function converts a date which has been entered as text to date value but in number format

Syntax

=DATEVALUE(DATE_TEXT)

DATE_TEXT

Date which needs to be converted from text to date format

Example

Use DATE function to calculate dates

Text Date	Result	Formula
05/05/2017	42860	=DATEVALUE(A15)
02/12/2018	43436	=DATEVALUE(A16)
01/03/2015	42064	=DATEVALUE(A17)
02/08/2014	41853	=DATEVALUE(A18)

Exercise

Use DATEVALUE function to calculate dates and then change the format of the result into Date format

Text Date	Result
31/12/2015	31/12/2015
3/01/2010	03/01/2010
11/12/2015	11/12/2015
01/01/2014	01/01/2014



TODAY

What does it do?

This function returns today's date in date format

Syntax

=TODAY()



No paramter

No paramter is required for this function

Example

Identify today's date

Result	Formula
13/03/2026	=TODAY()
08/03/2026	=TODAY()-5

Exercise

Calculate number of days between today's date and date in Col A

Col A	Result
31/12/2015	3725
03/01/2010	5913
11/12/2015	3745
01/01/2014	4454



NOW

What does it do?

This function returns current date and time in excel date and time format

Syntax

=NOW()

No paramter

No paramter is required for this function

Example

Identify today's date and time

Result	Formula
13/03/2026 9:42	=NOW()
08/03/2026 9:42	=NOW()-5

Exercise

Use NOW() to calculate values in COL A

Col A	Formula
Current date and time	13/03/2026 9:42
Date and Time 7 days ago	06/03/2026 9:42



DAY

What does it do?

This function returns the day of the month i.e. 1, 2 ..31 in numeric format. Do note that at times, you may numeric to get the date in numeric format

Syntax

=DAY(SERIAL_NUMBER)



Serial Number

Date from which you want to extract the date part

Example

Identify today's date and time

Col A	Result	Formula
13/03/2026 9:42	13	=DAY(A16)
08/03/2026 9:42	8	=DAY(A17)

Exercise

Use NOW() to calculate values in COL A and then use DAY function to extract date part

Col A	Results from NOW function	Results from D
Current date and time	13/03/2026 9:42	13
Date and Time 7 days ago	06/03/2026 9:42	6



/ have to change the format of the cell to

ate function

YEAR

What does it do?

This function returns the year month in a date in a number format. Do note that at times, you may have to use the YEAR function in a numeric format to get the date in numeric format

Syntax

=YEAR(SERIAL_NUMBER)

Serial Number

Date from which you want to extract the year part

Example

Identify today's date and time

Col A	Result	Formula
29/03/2017 12:11	2017	=YEAR(A16)
22/02/1951	1951	=YEAR(A17)

Exercise 1

Use NOW() to calculate values in COL A and then use YEAR function to extract year

Col A	Results from NOW function	Results from YEAR function
Current date and time	13/03/2026 9:42	2026
Date and Time 365 days ago	13/03/2025 9:42	2025

Exercise 2

Use YEAR, MONTH, DAY and DATE functions to get dates in COL A from Row 30 onwards. Row 29 has been used as a starting point

Date	Results	Formula
15/01/2016	Use this date to calculate dates from Row 30 onwards	
15/02/2016	15/02/2016	=DATE(YEAR(A29),MONTH(A30),DAY(A30))
15/03/2016	15/03/2016	
15/04/2016	15/04/2016	
15/05/2016	15/05/2016	
15/06/2016	15/06/2016	
15/07/2016	15/07/2016	
20/08/2017	15/08/2016	



o change the format of the cell to

ear function
6
5

n done as an example

3),MONTH(A28)+1,DAY(A28))

WEEKDAY

What does it do?

This function returns a number between 1 and 7 depending upon the day of the week. Default return is - Saturday which will be 7. Do note that at times, you may have to change the format of the cell to numeric

Syntax

=WEEKDAY(SERIAL_NUMBER, [return_type])

Serial Number

Date from which you want to extract the year part

return_type

Default settings i.e. if you do not mention this, Sunday would be 1, Monday 2 and so on. If you select it as

Example

Identify today's weekday number basis different return_type value

Col A	Result	Formula
13/03/2026	6	=WEEKDAY(A20)
13/03/2026	5	=WEEKDAY(A21,2)
13/03/2026	4	=WEEKDAY(A22,3)
13/03/2026	2	=WEEKDAY(A23,14)

Exercise 1

Use WEEKDAY() to calculate weekdays of dates mentioned in 'Date' column using 'Return Value'

Dates	Return Value	Results
01/01/2017	1	01/01/1900 0:00
01/01/2017	2	07/01/1900 0:00
01/01/2017	3	06/01/1900 0:00

Exercise 2

If any of the dates in the column below falls on Tuesday, then change that date to Friday. Use Weekday function. 1st has been done as an example

Dates	Weekday	New Date
28/03/2017	3	24/03/2017
29/03/2017	4	29/03/2017
21/03/2017	3	17/03/2017
01/01/2016	6	01/01/2016
01/01/2017	1	01/01/2017
01/01/2015	5	01/01/2015
01/01/2014	4	01/01/2014
01/01/2013	3	28/12/2012



Sunday would be 1, Monday 2...so on till
: to get the date in numeric format

: 2, Monday will be given 1, Tuesday 2 and so on.

unction to check weekdays with return value 1. You will

MINUTE

What does it do?

This function returns the minutes between 0 and 59

Syntax

=MINUTE(SERIAL_NUMBER)

Serial Number

Date or time from which you want to extract minutes

Example

Identify today's weekday number basis different return_type value

Col A	Result	Formula
13/03/2026 9:42	42	=MINUTE(A15)
10:15 am	15	=MINUTE(A16)
10:20 pm	20	=MINUTE(A17)

Exercise

Use MINUTE() to find out minutes from Col A

Col A	Results
31/01/2017 0:50	50
01/01/2017 0:01	1
12:45 am	45



HOUR

What does it do?

This function returns the hour from 0 to 23

Syntax

=HOUR(SERIAL_NUMBER)



Serial Number

Date or time from which you want to extract the hour

Example

Identify today's weekday number basis different return_type value

Col A	Result	Formula
13/03/2026 9:42	9	=HOUR(A15)
10:15 am	10	=HOUR(A16)
11:59 pm	23	=HOUR(A17)

Exercise

Use HOUR() to find out hour from Col A

Col A	Results
31/01/2017 0:50	0
01/01/2017 23:01	23
5:45 am	5



VLOOKUP

What does it do?

This function is used to lookup a particular value in a table and then return data from a specific column

Syntax

=VLOOKUP(LookupValue, Table_Array, Column_Index_Num, [Range_Lookup])

LookupValue

The value which needs to be looked for in a particular table

Table_Array

The table in which we are checking for this value to be present and from which we want to retrieve data. You fix the range. So when you copy formula from one cell to another, the range does not update automatically.

Column_Index_Num

If the value is present in the table, then which column consists of the value we want the function to return.

Range_Lookup

It is optional - it has 2 values - TRUE for approximate match and FALSE (which is default value) for exact match.

Example 1

Below in the **Lookup Table**, we have cost for different parts for different car makers. Use VLOOKUP() to find the cost of a part for a specific car maker.

Lookup Table	Mercedes	BMW	Audi
GearBox	\$2,500	\$2,000	\$3,000
Engine	\$5,000	\$4,999	\$5,200
Steering	\$1,250	\$1,302	\$1,100
Ignition	\$250	\$300	\$200
CYHead	\$300	\$290	\$310

Maker	Spare	Cost	Formula
Mercedes	Ignition	£250	=VLOOKUP(B31,\$A\$24:\$B\$28,2,FALSE)
BMW	GearBox	£2,000	=VLOOKUP(B32,\$A\$24:\$C\$28,3,FALSE)
Audi	Engine	£5,200	=VLOOKUP(B33,\$A\$24:\$D\$28,4,FALSE)
Audi	Steering	£1,100	=VLOOKUP(B34,\$A\$24:\$D\$28,4,FALSE)
Mercedes	Ignition	£250	=VLOOKUP(B35,\$A\$24:\$B\$28,2,FALSE)
Mercedes	CYHead	£300	=VLOOKUP(B36,\$A\$24:\$B\$28,2,FALSE)
BMW	GearBox	£2,000	=VLOOKUP(B37,\$A\$24:\$C\$28,3,FALSE)
BMW	Engine	£4,999	=VLOOKUP(B38,\$A\$24:\$C\$28,3,FALSE)

Example 2

Below are 3 tables, the first one consists of costs of 4 items, the second one consists of its sales price. Use the 3rd table and then use Units and the difference between Sales Price and Cost to calculate profit or loss.

Table 1

Table 2

Item Name	Cost
Pen	\$5.10
Pencil	\$1.10
Notepad	\$1.50
Ink	\$2.50

Item Name	Sales Price
Pen	\$7.50
Pencil	\$1.15
Notepad	\$2.50
Ink	\$3.75

Table 3					
Item Name	Units	Cost	Sales Price	Total Profit	For
Pen	500	\$5.10	\$7.50	\$1,200.00	=VLOOKUP(A5
Pencil	100	\$1.10	\$1.15	\$5.00	=VLOOKUP(A5
Notepad	250	\$1.50	\$2.50	\$250.00	=VLOOKUP(A5
Ink	100	\$2.50	\$3.75	\$125.00	=VLOOKUP(A5
Eraser	990	#N/A	#N/A		=VLOOKUP(A5
Total Profit = (Sales Price-Cost)*Units					

Since Eraser was not present in Table 1 and in Table 2, excel returns an error #N/A implying that the data we are searching for is not present

Exercise 1

Below are 3 different tables, in the first table we have employees and their designation, in the second table we have their names and department. Use Vlookup() to fill all these details in the 4th table

Table 1	
E Name	Designation
Ben	AVP
Ken	Analayst
Rachel	AVP
Katie	Analyst
Helen	Analyst

Table 2	
E Name	Salary
Ben	\$85,000
Ken	\$60,000
Rachel	\$65,000
Katie	\$50,000
Helen	\$45,000

Table 4			
E Name	Designation	Salary	Dept
Katie	Analyst	\$50,000	HR
Helen	Analyst	\$45,000	IT
Rachel	AVP	\$65,000	IT
Ben	AVP	\$85,000	Analytics
Richard	#N/A	#N/A	#N/A
Ken	Analayst	\$60,000	Analytics

Exercise 2

Below in the **Lookup** table, we have cost for different parts for different car makers. Use VLOOKUP() to

Lookup Table	GearBox	Engine	Steering	Ignition	CYHead
Mercedes	\$2,500	\$5,000	\$1,250	\$250	\$300
BMW	\$2,000	\$4,999	\$1,302	\$300	\$290
Audi	\$3,000	\$5,200	\$1,100	\$200	\$310

Maker	Mercedes	BMW	Audi	Audi	Mercedes
Spare	CYHead	GearBox	Gearbox	Steering	Ignition
Costs	\$300	\$2,000	\$3,000	\$1,100	\$250



corresponding to that value

1. By pressing F2 key when you have selected your table, automatically

return

match of the value we are searching for in a table

get costs in front of items in the second table

- 1) The first parameter relates to spare part for which we want to get the costs
 - 2) The second part refers to the data range - so Mercedes spare part information is available in the 'LookUp table' - so that is why you select only first 2 columns - 1st one consists of the item name which we are searching for and the second one consists of its cost. For BMW that information is in the 3rd column, thus our data range consists of 3 columns for the second parameter
 - 3) The 3rd parameter refers to from which column we want to return the value, for Mercedes it would be 2nd column of the selected range, for BMW it would be 3rd and for Audi it would be 4th
 - 4) FALSE refers to us wanting an exact match and not an approximate one

use VLOOKUP function to put the cost and sales price in

Formula for Cost	Formula for Sales Price
=VLOOKUP(A54,\$A\$46:\$B\$49,2,FALSE)	=VLOOKUP(A54,\$D\$46:\$E\$49,2,FALSE)
=VLOOKUP(A55,\$A\$46:\$B\$49,2,FALSE)	=VLOOKUP(A55,\$D\$46:\$E\$49,2,FALSE)
=VLOOKUP(A56,\$A\$46:\$B\$49,2,FALSE)	=VLOOKUP(A56,\$D\$46:\$E\$49,2,FALSE)
=VLOOKUP(A57,\$A\$46:\$B\$49,2,FALSE)	=VLOOKUP(A57,\$D\$46:\$E\$49,2,FALSE)
=VLOOKUP(A58,\$A\$46:\$B\$49,2,FALSE)	=VLOOKUP(A58,\$D\$46:\$E\$49,2,FALSE)

Table we have their names and salary and in the 3rd one

Table 3	
E Name	Dept
Ben	Analytics
Ken	Analytics
Rachel	IT
Katie	HR
Helen	IT

get costs of items in the second table

Mercedes	BMW	BMW
Engine	GearBox	Engine
\$5,000	\$2,000	\$4,999



able in second column
he spare part name
ation is available in

edes that would be

HLOOKUP

What does it do?

This function is similar to VLOOKUP. In a VLOOKUP, we look at columns, whereas in HLOOKUP, we look at rows. It searches for a value in the first row of a table, and returns a value from the same column in that table.

Syntax

=HLOOKUP(LookupValue, Table_Array, Row_Index_Num, [Range_Lookup])

LookupValue

The value which needs to be looked for in a particular table

Table_Array

The table in which we are checking for this value to be present and from which we want to retrieve data. You fix the range. So when you copy formula from one cell to another, the range does not update automatically.

Row_Index_Num

If the value is present in the table, then which row consists of the value we want the function to return

Range_Lookup

It is optional - it has 2 values - TRUE for approximate match (which is default) and FALSE for exact match

Example

Below in the first table, we have sales commission for different cars. Use VLOOKUP() to get amount each

Car Type	Mercedes	BMW	Audi	VW	Ferrari
Commission	\$500	\$450	\$550	\$200	\$1,000

Sold By	Maker	Commission	Formula
Ben	Mercedes	£500	=HLOOKUP(B29,\$B\$25:\$F\$26,2,FALSE)
Ben	BMW	£450	=HLOOKUP(B30,\$B\$25:\$F\$26,2,FALSE)
Alex	Audi	£550	=HLOOKUP(B31,\$B\$25:\$F\$26,2,FALSE)
Gary	Audi	£550	=HLOOKUP(B32,\$B\$25:\$F\$26,2,FALSE)
Helen	Ferrari	£1,000	=HLOOKUP(B33,\$B\$25:\$F\$26,2,FALSE)
Helen	Ferrari	£1,000	=HLOOKUP(B34,\$B\$25:\$F\$26,2,FALSE)
Ken	BMW	£450	=HLOOKUP(B35,\$B\$25:\$F\$26,2,FALSE)
Arnold	Mercedes	£500	=HLOOKUP(B36,\$B\$25:\$F\$26,2,FALSE)

Exercise

Below in the **Lookup** table, we have cost for different parts for different car makers. Use HLOOKUP() to get the cost. It has been done as an example

Lookup Table	Mercedes	BMW	Audi
GearBox	\$2,500	\$2,000	\$3,000
Engine	\$5,000	\$4,999	\$5,200
Steering	\$1,250	\$1,302	\$1,100

Ignition	\$250	\$300	\$200
CYHead	\$300	\$290	\$310

Maker	Mercedes	BMW	Audi	Audi	Mercedes
Spare	CYHead	GearBox	Gearbox	Steering	Ignition
Costs	£300	£290	£310	£310	£300
	=HLOOKUP(B50,\$B\$42:\$B\$47,6,FALSE)				



at rows. The HLOOKUP function looks for

a. By pressing F2 key when you have selected your table,
natically

1 of the value we are searching for in a table

h employee will get based on car they sold

- 1) The first parameter relates to car for which we want to get the commission amount
 - 2) The second part refers to the data range - so car names are in the 1st row of table 1 and information is in 2nd row of table 1
 - 3) The 3rd parameter refers to from which row we want to return the value - in this case which consists of the commission amount
 - 4) FALSE refers to us wanting an exact match and not an approximate one

get costs of items in the second table. First one has

Mercedes	BMW	BMW
Engine	GearBox	Engine
£300	£290	£290

1

d the commission

it is the 2nd row

TRANSPOSE

What does it do?

The TRANSPOSE function copies data from vertical cells to horizontal or vice versa

Syntax

=TRANSPOSE(Range)

Range

Data which you want to change from vertical to horizontal or vice-versa

How to use Transpose function

The transpose range must be the same size as the original range.

The function needs to be entered as an array formula.

To enter an array formula you must first highlight all the cells where the formula is required.

Next type the formula, such as =TRANSPOSE(A1:A5).

Finally press Ctrl+Shift+Enter to confirm it.

If changes need to be made to the formula, the entire array has to be highlighted, the edits can then be

The curly braces " { } " surrounding the function indicate that it is an array formula

Example

Use TRANSPOSE function to change the cells from vertical to horizontal

Employee ID	Jan	Feb
E001	10	30
E002	40	50
E003	70	80

EID	E001	E002	E003
Jan	10	40	70
Formula EID	=TRANSPOSE(A24:A26)		
Formula Jan	=TRANSPOSE(B24:B26)		

Exercise

Use TRANSPOSE function to change the cells from vertical to horizontal for Employee ID and Feb using s

Employee ID	Jan	Feb
E001	10	30
E002	40	50
E003	70	80

EID	E001	E002	E003
Jan	10	40	70



made and the Ctrl+Shift+Enter used to confirm it.

teps mentioned from Row 12 onwards

MATCH

What does it do?

This function is used to locate the position of a lookup value in a row, column, or table.

Syntax

=MATCH (Lookupvalue, Lookup_Array, [Match_type])

LookupValue

The value which needs to be looked for in a particular column, row or table

Lookup_Array

The column, table or table in which we are checking for this value to be present

Match_Type

It is optional - it has 3 values --> -1 gives the smallest value position that is greater than or equal to value search and 1 gives the largest value that is less than or equal to the value we are searching for. Default

Example 1

Below in the **Lookup** table, we have cost for different parts for different car makers. Use VLOOKUP() to

Lookup Table	Mercedes	BMW	Audi
GearBox	\$2,500	\$2,000	\$3,000
Engine	\$5,000	\$4,999	\$5,200
Steering	\$1,250	\$1,302	\$1,100
Ignition	\$250	\$300	\$200
CYHead	\$300	\$290	\$310

Maker	Spare	Cost	Formula
Mercedes	Ignition	£250	=VLOOKUP(B31,\$A\$24:\$B\$28,2,FALSE)
BMW	GearBox	£2,000	=VLOOKUP(B32,\$A\$24:\$C\$28,3,FALSE)
Audi	Engine	£5,200	=VLOOKUP(B33,\$A\$24:\$D\$28,4,FALSE)
Audi	Steering	£1,100	=VLOOKUP(B34,\$A\$24:\$D\$28,4,FALSE)
Mercedes	Ignition	£250	=VLOOKUP(B35,\$A\$24:\$B\$28,2,FALSE)
Mercedes	CYHead	£300	=VLOOKUP(B36,\$A\$24:\$B\$28,2,FALSE)
BMW	GearBox	£2,000	=VLOOKUP(B37,\$A\$24:\$C\$28,3,FALSE)
BMW	Engine	£4,999	=VLOOKUP(B38,\$A\$24:\$C\$28,3,FALSE)

Example 2

Below are 3 tables, the first one consists of costs of 4 Items, the second one consists of its sales price. Use the 3rd table and then use Units and the difference between Sales Price and Cost to calculate profit or loss

Table 1	
Item Name	Cost
Pen	\$5.10
Pencil	\$1.10

Table 2	
Item Name	Sales Price
Pen	\$7.50
Pencil	\$1.15

Notepad	\$1.50
Ink	\$2.50

Notepad	\$2.50
Ink	\$3.75

Table 3					
Item Name	Units	Cost	Sales Price	Total Profit	For
Pen	500	\$5.10	\$7.50	\$1,200.00	=VLOOKUP(A5
Pencil	100	\$1.10	\$1.15	\$5.00	=VLOOKUP(A5
Notepad	250	\$1.50	\$2.50	\$250.00	=VLOOKUP(A5
Ink	100	\$2.50	\$3.75	\$125.00	=VLOOKUP(A5
Eraser	990	#N/A	#N/A		=VLOOKUP(A5
Total Profit = (Sales Price-Cost)*Units					

Since Eraser was not present in Table 1 and in Table 2, excel returns an error #N/A implying that the data we are searching for is not present

Exercise 1

Below are 3 different tables, in the first table we have employees and their designation, in the second table we have their names and department. Use Vlookup() to fill all these details in the 4th table

Table 1	
E Name	Designation
Ben	AVP
Ken	Analayst
Rachel	AVP
Katie	Analyst
Helen	Analyst

Table 2	
E Name	Salary
Ben	\$85,000
Ken	\$60,000
Rachel	\$65,000
Katie	\$50,000
Helen	\$45,000

Table 4			
E Name	Designation	Salary	Dept
Katie	Analyst	\$50,000	HR
Helen	Analyst	\$45,000	IT
Rachel	AVP	\$65,000	IT
Ben	AVP	\$85,000	Analytics
Richard	#N/A	#N/A	#N/A
Ken	Analayst	\$60,000	Analytics

Exercise 2

Below in the **Lookup** table, we have cost for different parts for different car makers. Use VLOOKUP() to

Lookup Table	GearBox	Engine	Steering	Ignition	CYHead
Mercedes	\$2,500	\$5,000	\$1,250	\$250	\$300

BMW	\$2,000	\$4,999	\$1,302	\$300	\$290
Audi	\$3,000	\$5,200	\$1,100	\$200	\$310

Maker	Mercedes	BMW	Audi	Audi	Mercedes
Spare	CYHead	GearBox	Gearbox	Steering	Ignition
Costs	\$300	\$2,000	\$3,000	\$1,100	\$250



we are searching, 0 gives the value for the exact
is 1

get costs in front of items in the second table

- 1) The first parameter relates to spare part for which we want to get the costs
- 2) The second part refers to the data range - so Mercedes spare part information is available of the 'LookUp table' - so that is why you select only first 2 columns - 1st one consists of the item which we are searching for and the second one consists of its cost. For BMW that information is in the 3rd column, thus our data range consists of 3 columns for the second parameter
- 3) The 3rd parameter refers to from which column we want to return the value, for Mercedes it would be 2nd column of the selected range, for BMW it would be 3rd and for Audi it would be 4th
- 4) FALSE refers to us wanting an exact match and not an approximate one

use VLOOKUP function to put the cost and sales price in

Formula for Cost	Formula for Sales Price
=VLOOKUP(A54,\$D\$46:\$B\$49,2,FALSE)	=VLOOKUP(A54,\$D\$46:\$E\$49,2,FALSE)
=VLOOKUP(A55,\$D\$46:\$B\$49,2,FALSE)	=VLOOKUP(A55,\$D\$46:\$E\$49,2,FALSE)
=VLOOKUP(A56,\$D\$46:\$B\$49,2,FALSE)	=VLOOKUP(A56,\$D\$46:\$E\$49,2,FALSE)
=VLOOKUP(A57,\$D\$46:\$B\$49,2,FALSE)	=VLOOKUP(A57,\$D\$46:\$E\$49,2,FALSE)
=VLOOKUP(A58,\$D\$46:\$B\$49,2,FALSE)	=VLOOKUP(A58,\$D\$46:\$E\$49,2,FALSE)

Table 3 we have their names and salary and in the 3rd one

Table 3	
E Name	Dept
Ben	Analytics
Ken	Analytics
Rachel	IT
Katie	HR
Helen	IT

get costs of items in the second table

Mercedes	BMW	BMW
Engine	GearBox	Engine
\$5,000	\$2,000	\$4,999



able in second column
he spare part name
ation is available in

edes that would be

SUM

What does it do?

This function returns the sum of the cells or range selected

Syntax

=SUM(Number 1, Number 2...)

Number 1

The first value to sum

Number 2 and so on

The second (and so on..) value to sum

Example 1

Use SUM function to get the sum of cells

Col A	Col B	Col C	Result	Formula
5	12	17	34	=SUM(A18:C18)
10	18	23	51	=SUM(A19,B19,C19)
20	100	999	1119	=A20+B20+C20

Example 2

Use SUM function to get the sum of cells

Row A	5	9	12	21
Row B	1	56	345	23
Row C	2	6	23	34
Row D	4	12	547	12
Total	12	83	927	90
Formula	=SUM(B24:B27)	=C24+C25+C26+C27	=SUM(D24,D25,D26,D27)	=SUM(E24:E25,E26:E27)

Example 3

Get the SUM of all cells below

Values to be summed			Total	Formula
5		3	16	=SUM(A34,A36,C34,C36,B35:B36)
	1			
1	2	4		

Note

If you want to sum the entire column I suppose, all you need to do is =SUM(I:I), it will select the entire column. Similarly, you can select the entire row as well. If you write, =SUM(41:42), It will return the sum of all numbers in the row.

Exercise 1

Use SUM function to calculate sum of the following values

Values to be summed		Result
1		
100		
0	200	500

Exercise 2

Use IF and SUM function to calculate sum of the following values if conditions meet. If the conditions do not meet, return 0. Use the following example

Condition to meet	Col B	Col C	Col D	Col E	Result
Col B > 1	5	1	2	4	12
Col C > 50	9	56	6	12	83
Col C > 500	12	345	23	547	0
Col E > 80	84.3	59.8	80.9	80.5	305.5
Col D > 35	9.8	0.2	38	80.9	128.9

Exercise 3

Use IF, AND and SUM functions to calculate sum of the following values if conditions meet. If the condition does not meet, return 0. Use the following example

First Condition	Second Condition	Col B	Col C	Col D	Col E
Col B > 1	Col D >= 2	5	1	2	4
Col C > 50	Col A > 5	9	56	6	12
Col C > 500	Col E > 500	12	345	23	547
Col E > 80	Col A > 80	84.3	59.8	80.9	80.5
Col D > 35	Col B >= 9.8	9.8	0.2	38	80.9

Exercise 4

Use IF, OR and SUM functions to calculate sum of the following values if either of the 2 conditions meet. If the condition does not meet, return 0. Use the following example

First Condition	Second Condition	Col B	Col C	Col D	Col E
Col B > 1	Col D >= 2	5	1	2	4
Col C > 50	Col A > 5	9	56	6	12
Col C > 500	Col E > 500	12	345	23	547
Col E > 80	Col A > 80	84.3	59.8	80.9	80.5
Col D > 35	Col B >= 9.8	9.8	0.2	38	80.9



Column I and give you the sum of the numbers in it.
Numbers in Rows 41 and 42.

ot meet, then return value 0. First has been done as an

=IF(B54>1,SUM(B54:E54),0)

is do not meet, then return value 0. First has been done

Result	
12	=IF(AND(C65>1,E65>=2),SUM(C65:F65),0)
83	
0	
305.5	
128.9	

the conditions do not meet, then return value 0. First

Result	
12	=IF(OR(C76>1,E76>=2),SUM(C76:F76),0)
83	
927	
305.5	
128.9	

SUMIF

What does it do?

This function returns the sum of the cells which meet a given criteria

Syntax

=SUMIF(Range, Criteria, [sum_range])

Range

Range of cells which we need the criteria to evaluate

Criteria

This refers to the condition which we want to check, it can be text, number, dates

Sum Range

This is optional, if not present, excel will consider numbers in the 'Range'. Otherwise, it specifies the

Example 1

Use SUMIF function to calculate salary of all employees in Analytics and then for Sales

Employee ID	Department	Designation	Date of Joining	Salary
E001	HR	AVP	01/01/2017	\$40,000
E002	Analytics	AVP	01/12/2016	\$75,000
E003	Sales	AVP	01/12/2016	\$30,000
E004	Analytics	AVP	01/11/2016	\$80,000
E005	HR	Analyst	01/10/2016	\$35,000
E006	Sales	Analyst	01/11/2016	\$25,000
E007	Analytics	Analyst	01/10/2016	\$60,000
E008	Sales	Analyst	01/11/2016	\$20,000
E009	IT	Analyst	01/10/2016	\$40,000
E010	IT	Analyst	01/10/2016	\$38,000

Salary for Analytics Group

Salary for Analytics Group

Total	Formula
\$215,000	=SUMIF(B22:B31,"Analytics",E22:E31)
\$78,000	=SUMIF(B22:B31,"IT",E22:E31)

Example 2

Use SUMIF function to calculate salary of all employees who joined before 1st Dec 2016

Employee ID	Department	Designation	Date of Joining	Salary
E001	HR	AVP	01/01/2017	\$40,000
E002	Analytics	AVP	01/12/2016	\$75,000
E003	Sales	AVP	01/12/2016	\$30,000
E004	Analytics	AVP	01/11/2016	\$80,000

E005	HR	Analyst	01/10/2016	\$35,000
E006	Sales	Analyst	01/11/2016	\$25,000
E007	Analytics	Analyst	01/10/2016	\$60,000
E008	Sales	Analyst	01/11/2016	\$20,000
E009	IT	Analyst	01/10/2016	\$40,000
E010	IT	Analyst	01/10/2016	\$38,000

Salary for employees who joined before 1st Dec'16	Total	Formula
	\$298,000	=SUMIF(D41:D50,"<01-Dec-2016",E41:E50)

Exercise 1

Use SUMIF function to calculate salary of all employees who are AVP

Employee ID	Department	Designation	Date of Joining	Salary
E001	HR	AVP	01/01/2017	\$40,000
E002	Analytics	AVP	01/12/2016	\$75,000
E003	Sales	AVP	01/12/2016	\$30,000
E004	Analytics	AVP	01/11/2016	\$80,000
E005	HR	Analyst	01/10/2016	\$35,000
E006	Sales	Analyst	01/11/2016	\$25,000
E007	Analytics	Analyst	01/10/2016	\$60,000
E008	Sales	Analyst	01/11/2016	\$20,000
E009	IT	Analyst	01/10/2016	\$40,000
E010	IT	Analyst	01/10/2016	\$38,000

	Result	Formula
Salary for Analytics Group	\$225,000	\$225,000

Exercise 2

Use SUMIF function to calculate salary of all employees joined on 1st Oct 2016

Employee ID	Department	Designation	Date of Joining	Salary
E001	HR	AVP	01/01/2017	\$40,000
E002	Analytics	AVP	01/12/2016	\$75,000
E003	Sales	AVP	01/12/2016	\$30,000
E004	Analytics	AVP	01/11/2016	\$80,000
E005	HR	Analyst	01/10/2016	\$35,000
E006	Sales	Analyst	01/11/2016	\$25,000
E007	Analytics	Analyst	01/10/2016	\$60,000
E008	Sales	Analyst	01/11/2016	\$20,000
E009	IT	Analyst	01/10/2016	\$40,000
E010	IT	Analyst	01/10/2016	\$38,000

	Result	Formula
Salary for Analytics Group	\$173,000	\$173,000



cells which need to be added if the condition holds

SUMPRODUCT

What does it do?

This function returns the sum after multiplying numbers in an array. It can also be used to count cells based on criteria.

Syntax

=SUMPRODUCT(Array1, [Array2], [Array3],...and so on)

Array1

Range of cells which first need to be multiplied and then summed

Array2

It is optional - the second array or range to multiply and then summed

Example 1

Use SUMPRODUCT function to calculate total cost faced by the superstore

Items	Perishable?	Cost	No. of Units
I001	NO	£1.50	100
I002	YES	£1.99	150
I003	YES	£2.05	200
I004	NO	£3.50	50
I005	YES	£4.90	400
I006	NO	£5.00	1,000

	Total	Formula
Total Cost	\$7,994	=SUMPRODUCT(C18:C23,D18:D23)

Example 2

Use SUMPRODUCT function to count total number of Perishable items

Items	Perishable?	Total Cost
I001	NO	£150
I002	YES	£299
I003	YES	£410
I004	NO	£175
I005	YES	£1,960
I006	NO	£5,000

	Total	Formula
Total Items	3	=SUMPRODUCT(--(B32:B37="YES"))
Cost for Perishable Items	\$2,669	=SUMPRODUCT(--(B32:B37="YES"),C32:C37)

Exercise 1

Use SUMPRODUCT function to calculate total cost faced by the superstore

Items	Perishable?	Sales	No. of Units
I001	NO	£3.10	500
I002	YES	£2.50	150
I003	YES	£10.00	100
I004	NO	£1.00	500
I005	YES	£4.00	200
I006	NO	£2.00	2,000

	Total	Formula
Total Sales	\$8,225	⌘8,225

Exercise 2

Use SUMPRODUCT function to count total number of non Perishable items

Items	Perishable?	Total Cost
I001	NO	£150
I002	YES	£299
I003	YES	£410
I004	NO	£175
I005	YES	£1,960
I006	NO	£5,000

	Total
Total Items	3
Cost Non Perishable Items	\$5,325



used on criteria provided

RAND

What does it do?

This function returns random numbers greater than 0 and less than 1. The numbers change on recalculation.

Syntax

=RAND()



No Parameter

No Parameter is required for this function.

Example

Use RAND() to generate some random numbers.

Random No.s	Formula
0.440381249	=RAND()
0.793357268	=RAND()
0.293048987	=RAND()
0.003497144	=RAND()
0.162906095	=RAND()
0.135972915	=RAND()



ition

RANDBETWEEN

What does it do?

This function returns random numbers between specified numbers

Syntax

=RANDBETWEEN(BOTTOM, TOP)

BOTTOM

Lower Limit

TOP

Upper Limit

Example

Use RAND() to generate some random numbers

Random No.s	Formula
67335	=RANDBETWEEN(0,100000)
90942	=RANDBETWEEN(0,100000)
40280	=RANDBETWEEN(0,100000)
2591	=RANDBETWEEN(0,5000)
1601	=RANDBETWEEN(0,5000)
1952	=RANDBETWEEN(0,5000)



ROUND

What does it do?

This function rounds the numbers to specified digits.

Syntax

=ROUND(Number,num_digits)

Number

Number which you want to round up

num_digits

Number of decimal places to which you want it to round up

Example

Use ROUND function reduce decimal places of the numbers

Numbers	Rounded	Formula
45.3976106	45.40	=ROUND(A18,2)
50.6546265	50.70	=ROUND(A19,1)
85.4139093	85.414	=ROUND(A20,3)
33.5893471	34	=ROUND(A21,0)

Exercise

Use ROUND function reduce decimal places of the numbers

Items	Result	Formula
81.373	81.37	81.373
7.9597	8.00	7.960
521.2578	521.26	521.258
4.699	4.70	4.699



